A 22nm 15.2µJ/Inference Speech Enhancement Accelerator with Contextual-Aware KV-Cache Management and Reconfigurable D-CIM Engine

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***Abstract*—This paper presents the first Transformer-based SE accelerator for wearable hearing aids. To fit the Transformer model onto extreme low-power devices, the accelerator integrates a contextual-aware KV cache management, a reconfigurable digital compute-in-memory engine, and bit-wise input feature storage. The prototype 22nm chip achieves 15.2µJ/inference and 8.04TOPS/W at 10MHz,0.5V, which is 16.6× higher than a prior non-Transformer SE accelerator. Meanwhile, it achieves a superior 3.34 PESQ on long-range speech contexts (>200 frames).**

***Index Terms*—Speech enhancement, Transformer, KV cache, Compute-in-memory, Low-power device.**

I. Introduction

Speech enhancement (SE) is the key processing operation for wearable hearing aids, which aims to improve the quality and intelligibility of noisy or distorted speech signals from environment. In each year, approximately 22.69 million hearing aids devices are required [1], contributing to a $13.97 billion annual market. In general, the SE needs to achieve at least 10dB signal-to-distortion ratio (SDR) enhancement to ensure the hearing impaired patients receive comprehensible speech. As shown in Fig. 1, the prior SE typically adopted neural network model using an encoder-decoder architecture with stateful layers, e.g., LSTM [2], GRU [3] [4], to memorize features in continuous audio frames. However, it is quite challenging for these SE models to achieve high enhancement quality for speech context with a long range, e.g. speech with >80 frames.

Recent Transformer-based SE (TSE) models [5] [6] demonstrate superior capability on capturing long-range frame dependencies, showing significant improved key performance metrics, i.e., perceptual evaluation of speech quality (PESQ) and short-time objective intelligibility (STOI) for speech enhancement. Unlike the Transformers for language or vision, e.g., LLM and ViT, TSE model divides the extracted feature from encoder into multiple frequency banks (FBs), with corresponding Transformer block capturing dependency across audio frames. Furthermore, an attention window scheme is adopted in TSE to reduce operations and KV cache size by only executing attention across the most recent frames. Previously, specific hardware accelerators have only been explored for GRU-based SE [3] [7], while an efficient accelerator for TSE is strongly desired to accommodate to the long-range SE demands.

Deploying TSE models on wearable hearing aids devices faces a few key challenges from the perspective of memory, utilization and computing efficiency. First, the large KV cache generated by Transformer under streaming frames poses significant challenges for the limited edge memory (Fig. 2(a)). However, a significant portion of FBs carry no contextual information, with the corresponding KV cache is redundant and can be skipped by adjusting the attention window length (AWL), e.g. 8.09MB at AWL=200 to 2.52MB at AWL=50. Second, the TSE workflow exhibits significant workload fluctuation due to the varying speech context length, leading to hardware under-utilization

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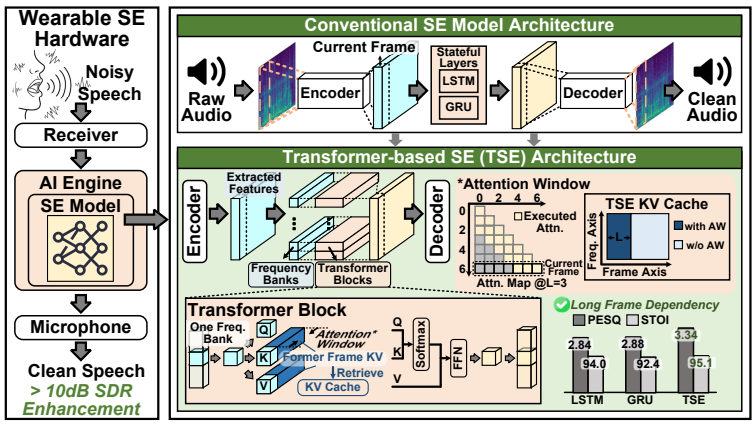


Fig. 1. Overview of speech enhancement (SE) workflow, the conventional and Transformer-based SE model architectures.

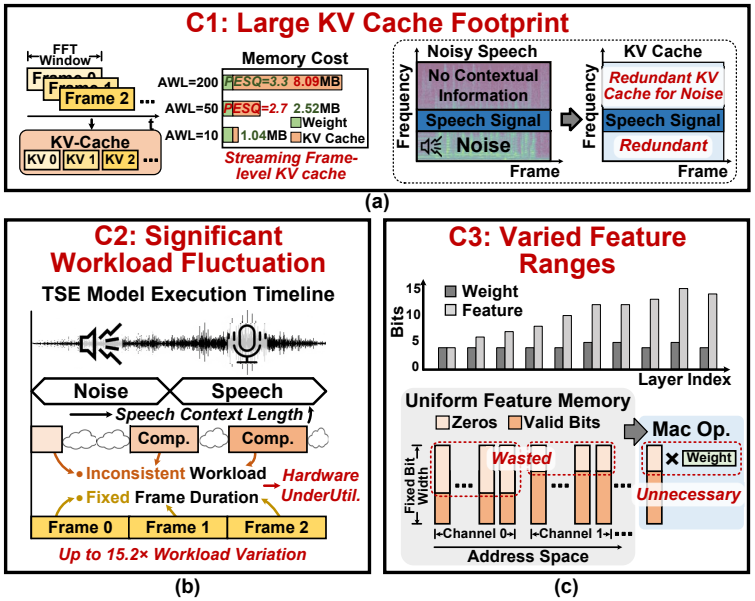


Fig. 2. Challenges for deploying Transformer model for hearing aids: (a) Large KV cache footprint, (b) Significant workload fluctuation, and (c) Varied input feature ranges.

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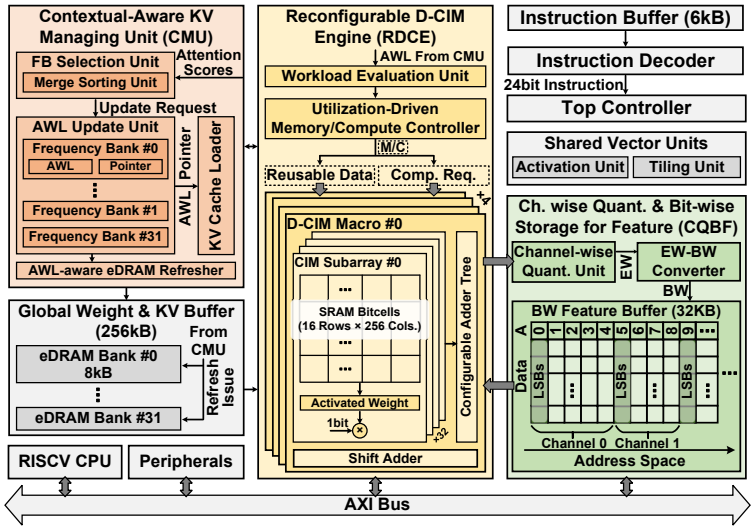


Fig. 3. Architecture overview of proposed Transformer SE accelerator.

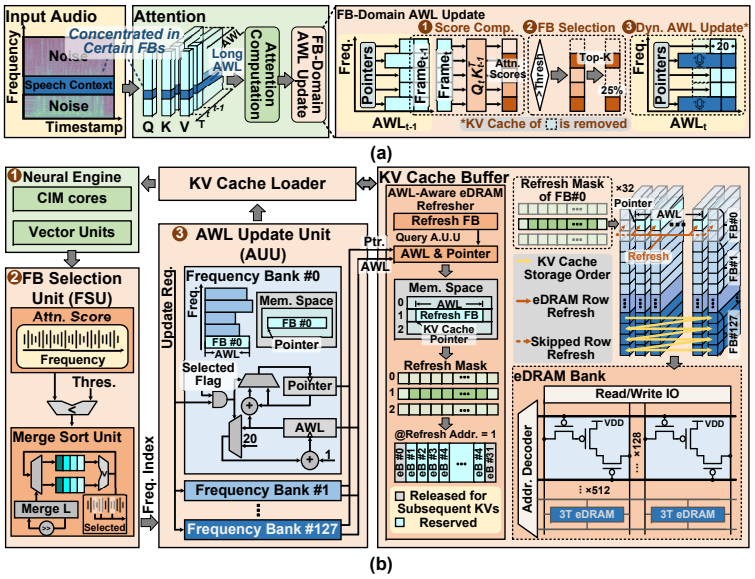


Fig. 4. (a) Contextual-aware KV cache management, (b) Design details CMU module.

(Fig. 2(b)). Since the Transformer operations are determined by the actual speech context, the large context length variation (up to 15.2× variation) generates inconsistent workloads within the fixed frame duration, e.g. 16ms. Third, the input features of TSE models have a much wider range of data values (require 4-16 bits quantization) compared to weights (4-5 bits), leading to low memory storage and computation efficiency (Fig. 2(c)). This variability stems from the inconsistent nature of audio input and is amplified by the intra-layer accumulations.

To address the above challenges, we present an efficient Transformer-based speech enhancement accelerator with following features: 1) a contextual-aware KV cache management with dynamic AWL adjustment based on runtime attention scores, 2) a reconfigurable digital compute-in-memory (D-CIM) engine with tunable memory-compute ratio for varying context length, and 3) a channel-wise mixed-precision quantization and efficient bit-wise (BW) storage for wide-range input features. With the above innovations, our accelerator achieves an inference efficiency of 15.2μJ/inference with 8.04TOPS/W computation efficiency, which is 16.6× higher than prior SE accelerator [3] with superior PESQ metric for long-range speech contexts of >200 frames.

II. Transformer SE Accelerator Overview

Fig. 3 shows the overview of chip architecture. It integrates a contextual-aware KV management unit (CMU), a reconfigurable D-CIM engine (RDCE), a channel-wise feature quantization unit with 32kB bit-wise feature buffers (CQBF), a 256kB global weight and KV cache buffer, a host RISC-V CPU and peripheral circuits. During the real-time operations, the CMU first identifies FBs with meaningful speech context based on runtime attention scores and updates AWL for each FB. To reduce memory area and cost, the weitht and KV cache buffer is organized with compact 3T eDRAM. An AWL-aware eDRAM refresher is designed to receive the updated AWL and skip the eDRAM refresh for the memory addresses with invalid KVs. The RDCE comprises four 16kB D-CIM macros with peripheral accumulation logics as the main computing cores. Based on online context workload evaluations, the memory-computation ratio of the RDCE can be reconfigured to enhance the hardware utilization. A channel-wise feature quantization unit is designed to quantize the computation results from RDCE to INT4-INT24 for each feature channel. The elements of each feature channel pass through the element-wise and bit-wise (EW-BW) converter, which transforms quantized EW features into BW and stores them into local BW feature buffer for subsequent computation.

III. Architecture Design Techniques

Fig. 4 (a) illustrates the workflow of our contextual-aware KV cache management. For real-time input audio, the speech context tends to be concentrated within a few certain FBs. These FBs are first identified by attention score computation (step 1) and Top-K selection (step 2), followed by dynamic AWL update across FB domains (step 3). The attention score, i.e. QK^T in attention operation, is computed between the current Frame$_t$ and previous Frame$_{t-1}$ for each FB by the D-CIM neural engine. Fig. 4(b) shows the detailed architecture of CMU. In the FB selection unit (FSU), a merge sort unit is utilized to sort attention scores and the indices of Top-K FBs are selected and sent to the following AWL update unit (AUU). The AWLs of selected FBs will be increased to include the current frame to maintain SE quality, while AWLs of the rest FBs are reset to a small length to drop the former KV cache with updated KV cache pointer. By such management, only FBs with contextual information (normally ∼25% FBs) are maintained using long AWL, while the rest redundant KV cache can be removed with no SE quality degradation.

The KV cache is stored by 32 on-chip 8kB eDRAM banks, which is designed using compact 3T bit cell for area efficiency. KV cache of each FB is stored in the same addresses across 32 eDRAMs and fill all 32 banks at one address before switching to the next. The AWL-aware eDRAM refresher refreshes eDRAM banks at the granularity of FB, with the KV cache pointer and AWL queried from AUU for the target refresh FB. A refresh mask is generated based on locations of valid KV cache to indicate the eDRAM banks require refresh, while

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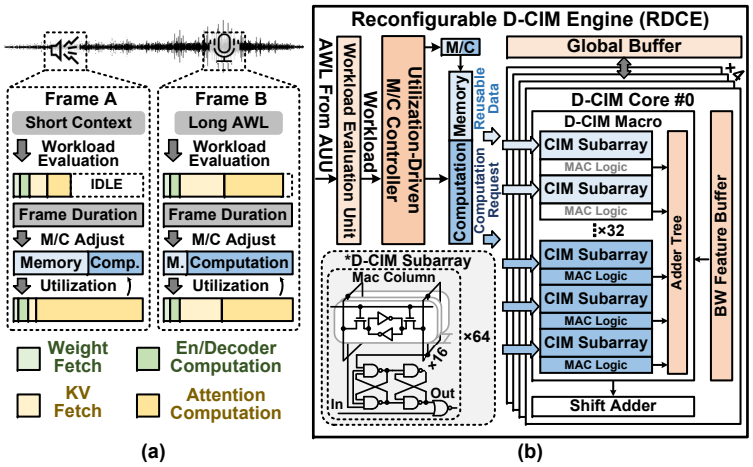


Fig. 5. (a) Reconfigurable D-CIM engine workflow under different input context, (b) Hardware design architecture of RDCE.

the rest banks are skipped and the corresponding memory space is released for subsequent KVs.

Fig. 5 shows the workflow and design details of our reconfigurable D-CIM engine. To accommodate to varying speech length, the workload of attention computation and corresponding KV fetching for the current frame is evaluated based on AWLs from the AUU. For frames without speech context, a utilization-driven memory/compute (M/C) controller increases the memory-compute ratio by dedicating more D-CIMs as memory to store reusable data, e.g., temporary computing results. Otherwise, the M/C controller reduces M/C ratio to enhance computation performance to realize better throughput. The reconfigurable D-CIM engine contains four 16kB D-CIM macros, each comprises 32 subarrays with 64 MAC columns and can be partitioned as either compute mode or memory mode with MAC logic disabled.

As shown in Fig. 6(a), to achieve efficient input feature storage and movement, the features are channel-wise quantized from INT32

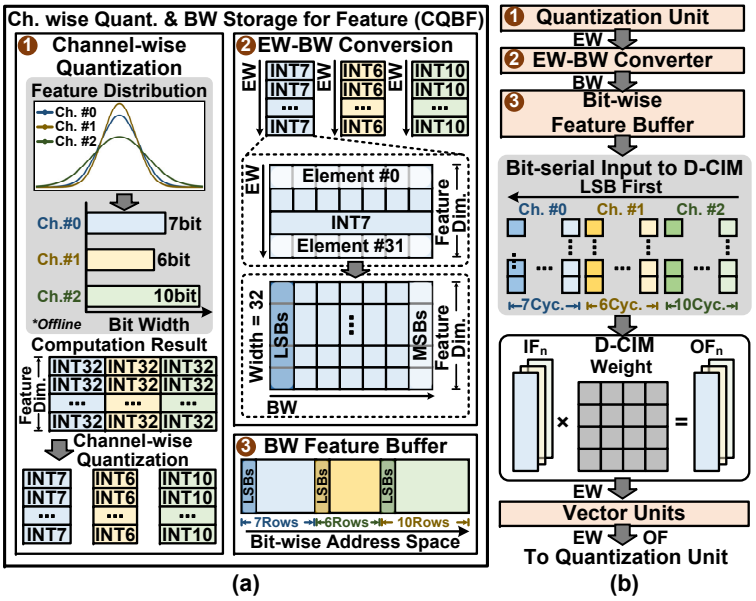
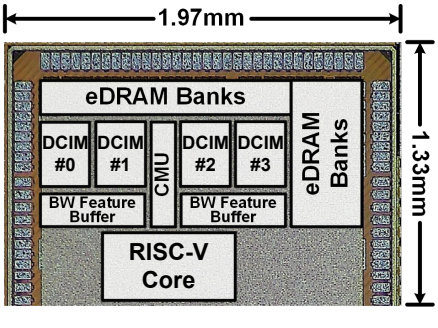


Fig. 6. (a) Channel-wise quantization with BW storage for input feature, (b) Computation dataflow of CQBF with D-CIM engine.



Technology	TSMC 22nm
Task	Speech Enhancement
Model Type	Transformer
Core Area	1.36 mm^2
Memory Size	4Mb eDRAM 64kB D-CIM 38kB SRAM
Data Precision	INT8 Weight INT4-24 Feature
Voltage	0.5-1.0V
Frequency	10-450MHz
Power	1.9mW
Throughput	125Frames/sec
Inference Efficiency	15.2μJ/Inference

Fig. 7. Chip photo and design specifications.

into mixed precision INT4-INT24 based on feature data distribution profiling (step 1). The quantized features of each channel are converted into BW format by grouping the bits from the same position of all elements (step 2), e.g., the first row of 32-bits BW channel #0 consists of LSBs from all 32 feature elements. The BW features are in-series stored into the BW feature buffer across the rows in an LSB-to-MSB order (step 3). As in the illustrated example, the features of channel #0 is quantized into INT7 and occupies 7 rows in the buffer, while the channel #2 is stored across 10 rows for INT10 quantization. During the computation illustrated in Fig. 6(b), the BW features are delivered into the D-CIMs in a bit-serial manner with the transfer cycles determined by the bit-precision of each channel. A vector unit is also designed to collect output features from D-CIMs and performs activations.

IV. Evaluation Results

Fig. 7 presents the chip photo and specifications. Our SE accelerator is fabricated on 22nm CMOS process and operates at 0.5V-1.0V with 10-450MHz. The peak inference efficiency is 15.2μJ/inference with only 1.9mW power at 0.5V, 10MHz, which is the lowest frequency to support real-time SE with a throughput of 125frames/sec (Fig. 8(a)). Our breakdown analysis shows the contextual-aware KV cache management, reconfigurable D-CIM engine and bit-wise feature storage contributes to the inference efficiency improvement by 2.17×, 1.21× and 1.43× respectively, for a total improvement of 3.75× (Fig. 8(b)).

Fig. 9(a) shows KV cache size reduction due to our contextual-aware KV management. With CMU, the average KV cache size is significantly reduced to 1.17MB, presenting a 6.32× reduction from the conventional 7.4MB KV cache with a fixed 200-frame AWL. It is worth to mention our context-aware AWL adjustment

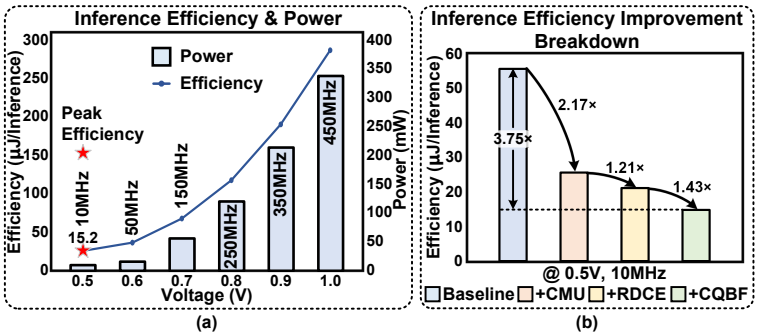


Fig. 8. (a) Measured power and inference efficiency, (b) Improvement breakdown.

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TABLE I
COMPARISON TABLE WITH PRIOR SOTA ACCELERATORS.

	ISSCC'23 [3]	ISSCC'25 [8]	JSSC'24 [9]	This Work
Technology	28	40	28	22
Speech Task	Speech Enhancement	BMI Speech Decoding	Speech to Text (ASR)	Speech Enhancement
Model Type	GRU/CNN	GRU	GRU/LSTM	Transformer/CNN
Core Area (mm^2)	0.81	2.22	9.58	2.15
Operation Voltage (V)	0.8-1.1	0.6-1.05	0.6-1.1	0.5-1.0
Clock Frequency (MHz)	2.5-20	2-200	1.25-100	10-450
Power (mW)	0.74 (0.8V,2.5MHz)	3.7 (0.6V,6.7MHz)	1.3 (0.6V,1.25MHz)	1.9 (0.5V,10MHz)
Mac Type	Multiplier-less PE Array	PE Array	PE Array	Reconfigurable Digital CIM
Data Precision	4bit Log-Scale	4/8bit	8bit Weight, 11bit Feature	8bit Weight, 4-24bit Feature
KV Cache Optimization	×	×	×	Contextual-Aware KV Management
Performance (GPOS)	0.09[a]	1.72[b]	1.35[a]	10.24
MACs/Frame (M)	0.71	6.86[c]	6.54[d]	39.6
Frames/sec	63	125	103	125
Computation Efficiency (TOPS/W)	0.484[e]	0.49[e]	1.04[e]	8.04[f]
STOI/PESQ	91.44/2.73	/	/	95.11/3.34

[a]Computed by (MACs/Frame) × (frame/sec) × 2 [b]Computed by (MAC size) × Frequency × 2 [c] Computed by performance ÷ (frames/sec) ÷ 2
[d]Assuming Input Feature Dimension = 40 [e]Computed by performance ÷ power [f]@0.5V, 10MHz, dense SE model by performance

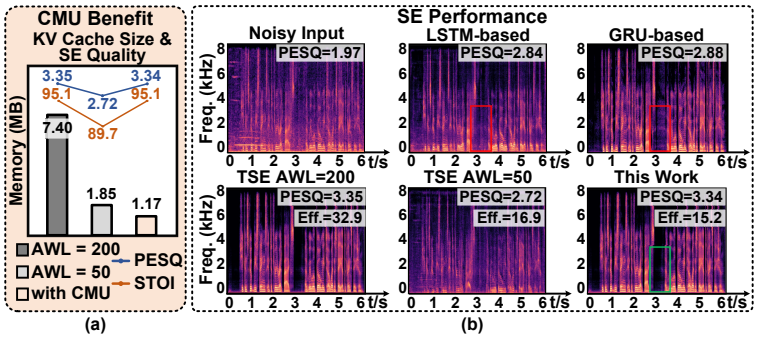


Fig. 9. (a) KV cache size under different attention AWL sizes, (b) SE performance of different SE models on an example audio segment.

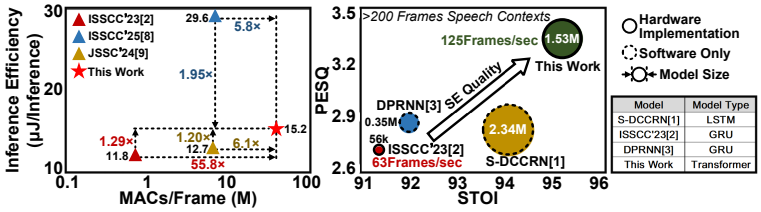


Fig. 10. Metric comparison with prior hardware accelerators and SE software models.

achieves no degradation on PESQ and STOI, while a fixed 50-frame AWL has obvious quality degradation. Fig. 9(b) provides a spectrogram example of real-time SE for a 6s noisy speech input (750 frames) from the VoiceBank+DEMAND dataset [10]. Compared with conventional LSTM/GRU-based SE models, our accelerator achieves notable better SE quality in speech-intensive scenarios with a PESQ of 3.34, while a higher 15.2μJ/inference efficiency is obtained compared with conventional fixed AWL schemes.

Table 1 presents the comparison table with prior SOTA accelerators. Due to our comprehensive innovations in KV cache management, reconfigurable D-CIM and bit-wise feature storage, an efficiency of 8.04TOPS/W is achieved for dense model computation. Compared with the prior SE accelerator [3], our chip can perform Transformer-based speech enhancement with 55.8× more MACs/frame using 1.29× inference energy (15.2 μJ/inference). When running TSE model at 2× frame throughput than [3], our chip achieves SE quality of 3.34 PESQ and 95.1 STOI on the VoiceBank+DEMAND dataset for long-range speech contexts with >200 frames, notably outperforming [3].

As shown in Fig. 10, compared with prior algorithm-level LSTM [2] or GRU [4] SE models, our TSE model achieves higher SE quality with only 1.35M parameters, which is applicable for edge wearable SE devices. We also compare our chip with prior GRU-based accelerators. Compared with [8] and [9], our chip achieves 16.4× and 7.7× higher computation efficiency while processing 5.8× and 6.1× more MACs/frame. Our design offers a promising SE solution for long-range speech contexts with >200 frames for wearable hearing aids.

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